

achieve particle energy levels in the range of hundreds of MeV to GeV is so gigantic as to make it impractical as a weapon. The accelerators capable of generating weapon-grade particle energies need to be driven by electrical power in the range of tens to hundreds of megawatts, almost equivalent to the capacity of a power sub-station.

Another limitation of a particle beam weapon arises out of the beam spreading, called blooming. There is thermal blooming and electrostatic blooming. Due to blooming, particle energy that would otherwise be focused on the target gets spread out.

Thermal blooming is present in both charged-particle and neutral-particle beams. It occurs when particles bump into one another under the effects of thermal vibration and also when they bump into air molecules.

Electrostatic blooming occurs only in charged-particle beams and is due to mutual repulsion of the charged particles. In the case of neutral-particle beams therefore, the beam spreading is due to thermal blooming only.

Blooming leads to an increasing beam size as it travels the distance to the target, thereby decreasing the beam intensity and consequently the ability to inflict damage. In other words, it reduces operational range. It may be mentioned here that electrostatic blooming is present in charged-particle beams. Neutral-particle beams are not affected by electrostatic blooming as absence of charge means no mutual repulsion and consequent beam spreading.

Yet another problem with particle beam weapons is their complicated beam control, more so in the case of neutral-particle beam weapons for exoatmospheric applications. One of the reasons for this is deflection caused by earth's magnetic field. The other reason, applicable to neutral-particle beams,

is the difficulty in sensing deviation of the beam from intended path to apply correction if required.

Effects of particle beam weapons

Due to the form of energy propagated by a particle beam weapon, the mechanism by which it destroys a target is different from those of other types of directed-energy weapons, including high-energy laser weapons and high-power microwaves. Both charged-particle and neutral-particle beam weapons generate their destructive power by accelerating subatomic particles or atoms to near speed of light and focusing these high-energy particles into a beam whose energy is the aggregate kinetic energy of the particles. When concentrated into a beam, it can melt or fracture the material on interaction with the target. Target destruction in the case of a particle beam weapon occurs by kinetic penetration due to sub-atomic particles moving at speed of light, thermal damage and disruption of atomic bonds of the target.

Particle beam weapons, like other forms of directed-energy weapons, affect their targets through either a soft kill or hard kill, depending on conditions such as distance to target, power generated by the weapon and target hardening level. In the case of soft-kill damage, the effect of attack is to deny or degrade the operation of the target platform, or even inflict partial damage. Some examples include disrupting electronics of a guided missile, forcing it to miss the intended target, or damaging visible, infrared and microwave sensors on board the target platform. Though the soft-kill damage causes temporary loss of function, it can seriously compromise mission success. Hard kill relates to a permanent physical damage to the structure of the target.

To be continued...

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